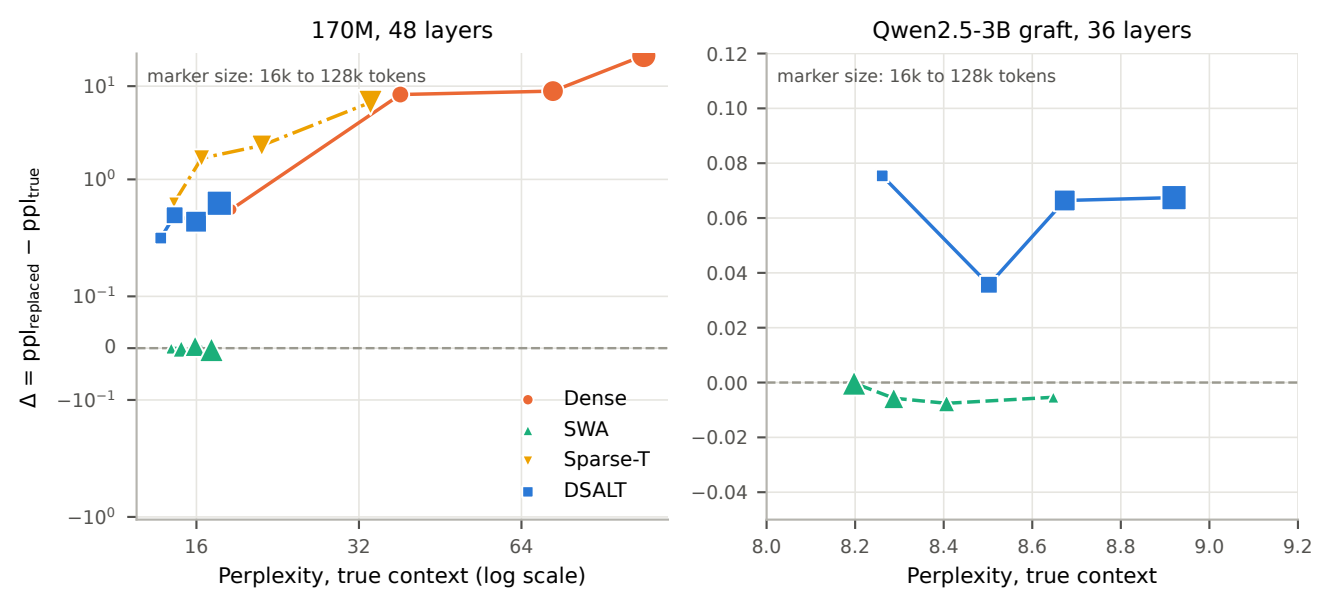
170M, 48 layers
marker size: 16k to 128k tokens
10^1
10^0
10^{-1}
0
-10^{-1}
-10^0
$\Delta = \mathrm{ppl}_{\mathrm{replaced}} - \mathrm{ppl}_{\mathrm{true}}$
Dense
SWA
Sparse-T
DSALT
16
32
64
Perplexity, true context (log scale)
Qwen2.5-3B graft, 36 layers
marker size: 16k to 128k tokens
0.12
0.10
0.08
0.06
0.04
0.02
0.00
−0.02
−0.04
8.0
8.2
8.4
8.6
8.8
9.0
9.2
Perplexity, true context